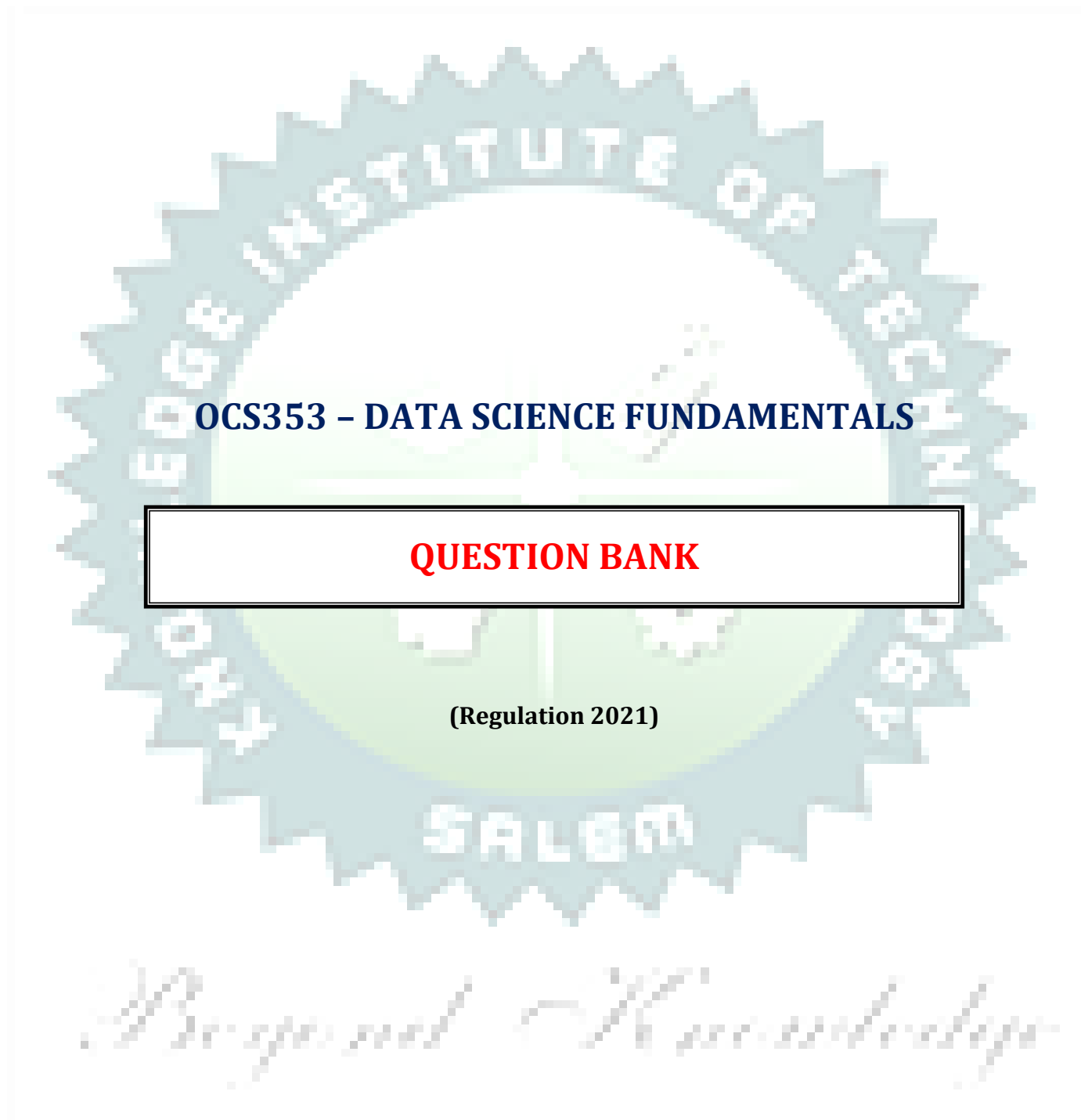




UNIT I

INTRODUCTION

PART A QUESTIONS AND ANSWERS

1. What is Data Science?

Data science is the study of data to extract meaningful insights for business. It is a multidisciplinary approach that combines principles and practices from the fields of mathematics, statistics, artificial intelligence, and computer engineering to analyze large amounts of data. This analysis helps data scientists to ask and answer questions like what happened, why it happened, what will happen, and what can be done with the results.

2. Differentiate between Data Science and Big Data.

Data Science	Big data
Data Science is an area	Big Data is a technique to collect, maintain information. and process huge information.
It is about the collection, processing. analyzing, and utilizing of data in various operations. It is more conceptual.	It is about the collection, processing. analyzing, and utilizing of data in various operations. It is more conceptual.
It is about extracting vital and valuable information from a huge amount of data	It is a superset of Big Data as data science consists of Data scrapping, cleaning, visualization, statistics, and many more techniques.
It is a sub-set of Data Science as mining activities which is in a pipeline of Data science	It is a superset of Big Data as data science consists of Data scrapping, cleaning, visualization, statistics, and many more techniques.
It is a sub-set of Data Science as mining activities which is in a pipeline of Data science	It is a field of study just like Computer Science, Applied Statistics, or Applied Mathematics
It is a technique for tracking and discovering trends in complex data sets.	It is a technique for tracking and discovering trends in complex data sets. The goal is to make data more vital and usable i.e. by extracting only important information from the huge data within existing traditional aspects.
It broadly focuses on the science of the data.	It is more involved with the of handling voluminous data.

3. Mention the characteristics of Big Data.

The characteristics of big data are often referred to as the three Vs:

- Volume-How much data is there?
- Variety - How diverse are different types of data?
- Velocity - At what speed is new data generated?
- Often these characteristics are complemented with a fourth V, Veracity - How accurate is the data?

4. What are the benefits and uses of Data Science

- Commercial companies in almost every industry use data science to gain insights into their customers, processes, staff, completion, and products.

- Governmental organizations are also aware of data's value.
- Nongovernmental organizations (NGOs) are also no strangers to using data. They use it to raise money and defend their causes.
- Universities use data science in their research but also to enhance the study experience of their students.

5. What are the main categories of Data?

- Structured
- Unstructured
- Natural language
- Machine-generated
- Graph-based
- Audio, video, and images
- Streaming

6. Define Structured Data.

Structured data is data that depends on a data model and resides in a fixed field SQL, or Structured Query Language, is the preferred way to manage and query data that resides in databases within a record

7. Define Unstructured Data with an example.

Unstructured data is data that isn't easy to fit into a data model because the content is context-specific or varying. One example of unstructured data is regular email. Although email contains structured elements such as the sender, title, and body text, it's a challenge to find the number of people who have written an email complaint about a specific employee because so many ways exist to refer to a person.

8. What is a machine generated data?

Machine-generated data is information that's automatically created by a computer, process, application, or other machine without human intervention

9. What is a graph based data?

In graph theory, a graph is a mathematical structure to model pair-wise relationships between objects. Graph or network data is, in short, data that focuses on the relationship or adjacency of objects. The graph structures use nodes, edges, and properties to represent and store graphical data. Graph-based data is a natural way to represent social networks, and its structure allows to calculate specific metrics such as the influence of a person and the shortest path between two people. Examples of graph-based data can be found on many social media websites.

10. Mention the steps of Data Science Process.

1. Setting the research goal
2. Retrieving data
3. Data preparation
4. Data exploration
5. Data modeling or model building
6. Presentation and automation

11. What are the steps needed in creating a project charter?

- A clear research goal
- The project mission and context
- How the developer going to perform the analysis
- What resources are expected to be used
- Proof that it's an achievable project, or proof of concepts
- Deliverables and a measure of success a timeline

12. Define Data Cleansing.

Data cleansing is a subprocess of the data science process that focuses on removing errors in data so that it becomes a true and consistent representation of the processes it originates from.

13. What are the different ways of combining Data?

Perform two operations to combine information from different data sets.

- 1. The first operation is joining:** enriching an observation from one table with information from another table.
- 2. The second operation is appending or stacking:** adding the observations of one table to those of another table

14. Define Pareto Diagram.

A Pareto diagram is a combination of the values and a cumulative distribution. Simple graphs can be combined which is called as Pareto diagram or 80-20 diagram.

15. What is the purpose of brushing and linking?

Brushing and linking combines and link different graphs and tables (or views) so changes in one graph are automatically transferred to the other graphs.

16. Define Dummy Variables.

Variables can be turned into dummy variables. Dummy variables can only take two values: true(1) or false(0). They're used to indicate the absence of a categorical effect that may explain the observation.

17. How to fix capital letter mismatches?

Capital letter mismatches are common. Most programming languages make a distinction between "Brazil" and "brazil". The problem is solved by applying a function that returns both strings in lowercase, such as `lower()` in Python. `"Brazil".lower()` `"brazil".lower()` should result in true. two operations to combine information from different data sets.

PART B QUESTIONS

1. Explain in details the Facets of Data?
2. State the differences between Big data and Data science. Mention the Benefits and uses of data science and big data
3. Explain the data science Process and its steps in details.
4. Discuss about defining research goals and the steps involved in creating a project charter?
5. Explain in detailed about Retrieving data in Data science?
6. Explain about cleaning, integrating and transforming data in detail?
7. Explain about data Exploration which discuss about graphs?
8. Explain about basic statistical description of data?
9. Explain standard deviation and its formula with an example
10. Explain Datamining?
11. Explain in detail about Text Mining with the steps involved in it?

UNIT II

DATA MANIPULATION

PART A QUESTIONS AND ANSWERS

1. What a Python Integer contains?

A single integer in Python actually contains four pieces:

- ob refent, a reference count that helps Python silently handle memory allocation and deallocation
- ob type, which encodes the type of the variable
- ob size, which specifies the size of the following data members
- ob digit, which contains the actual integer value that we expect the Python variable to represent

2. Mention Fixed-Type Arrays in Python with an example.

Python offers several different options for storing data in efficient, fixed-type data buffers. The built-in array module) can be used to create dense arrays of a

uniform type:

```
import array
```

```
L= list(range(10)) Aarray.array('i', L) A Out[]: array('i', [0, 1, 2, 3, 4, 5, 6, 7, 8, 9])
```

Here 'i' is a type code indicating the contents are integers.

3. How to Create Arrays from Python Lists?

First, we can use np.array to create arrays from Python lists.

```
Ex: np.array([1, 4, 2, 5, 3])
```

Here 'i' is a type code indicating the contents are integers

4. How does Array indexing in NumPy takes place?

In one-dimensional array accessing the ith value(counting from zero) is done by specifying the desired index in square brackets, just as with Python lists. In a array, accessing the ith value (counting from zero) is done

```
In [] = x1
```

```
Out []: array ([5, 0, 3, 3, 7, 91])
```

```
In []: x1[0]
```

```
Out []: 5
```

```
Tn []: x * 1[4]
```

```
Out []: 7
```

In a multidimensional array, access the items using a comma-separated tuple of indices:

```
In []:*2
```

```
Out[]: array [ [3, 5, 2, 4][7, 6, 8, 8], [1, 6, 7, 7] ]]
```

```
In[]:* 2[, 0,0]
```

```
Out[]: 3
```

5. Mention the syntax of Array Slicing in NumPy.

The NumPy slicing syntax follows that of the standard Python list, to access a slice of an array x, uses the following syntax

x[start: stop: step]

6. How Reshaping of Arrays is done?

Reshaping of arrays is the most flexible way of doing with the reshape() method. For example, To put the numbers 1 through 9 in a 3 * 3 grid, the following can be done:

```
In[38] grid = np.arange(1, 10).
```

```
reshape((3, 3)) print(grid)
```

```
[[1 2 3]]
```

```
[4 5 6]
```

7. low concatenation and splitting of arrays in Numpy is done?

Concatenation, or joining of two arrays in NumPy, is primarily done through the routines np.concatenate, np.vstack, and np.hstack

- The opposite of concatenation is splitting, which is implemented by the functions np.split, np.hsplit, and np.vsplit

8. Mention the rules of Broadcasting.

Rules of Broadcasting: Broadcasting in NumPy follows a strict set of modes determine the interaction between the two arrays:

Rule 1: If the two arrays differ in their number of dimensions, the shape of the one with fewer dimensions is padded with ones on its leading (left) side

Rule 2: If the shape of the two arrays does not match in any dimension, the array with shape equal to 1 in that dimension is stretched to match the other shape.

Rule 3: If in any dimension the sizes disagree and neither is equal to 1, error is raised.

9. Define Fancy Indexing.

Fancy indexing is conceptually simple: it means passing an array of indices to access multiple array elements at once.

10. Differentiate between Pandas and NumPy.

PANDAS	NUMPY
To work on Tabular data, the pandas module is preferred	To work on Numerical data, the numpy module is preferred.
The powerful tools of pandas are Data frame and Series.	The powerful tool of numpy is Arrays
Pandas consume more memory.	Numpy is memory efficient.
Indexing of the pandas series is very slow as compared to numpy arrays.	Indexing of numpy arrays is very fast.
Pandas offer a 2d table object called DataFrame.	Numpy is capable of providing multi-dimensional arrays.

11. Define Pandas Objects.

Pandas objects can be thought of as enhanced versions of Numpy structured arrays in which the rows and columns are identified with labels rather than simple integer indices.

12.Explain about Pandas Data structures

The three fundamental Pandas data structures are the Series DataFrame, and Index. A Pandas Series is a one-dimensional array of indexed data. The Series access with the values and index attributes

A DataFrame is an analog of a two-dimensional array with both flexible row indices and flexible column names.

The Index object is an interesting structure in itself, and it can be thought of either as an immutable array or as an ordered set technically a multiset, as Index objects may contain repeated values.

13.Name the ways Constructing of DataFrame objects is done.

A Pandas DataFrame can be constructed in a variety of ways.

- From a single Series object.
- From a list of dicts.
- From a dictionary of Series objects.
- From a two-dimensional NumPy array.
- From a NumPy structured array.

14.Define Indexers loc, iloc, and ix.

- The loc attribute allows indexing and slicing that always references the explicit Index.
- The iloc attribute allows indexing and slicing that always references the implicit Python-style index
- A third indexing attribute, ix, is a hybrid of the two, and for Series objects is equivalent to standard []-based indexing.

15.How to handle missing data in a table or DataPromat

A member of whemes have been developed to indicate the data in a table or Duts Frame Generally, they revolve around one of two strategies

- 1) Using a mask that globally indicates missing values
- 2) Choosing a sentinel value that indicates a missing entry.

16. State NoN and None in Pandas.

NaN and None both have their place, and Pandas is built to handle the theof them nearly interchangeably, converting between them where appropriate

```
In[10]: pd.Series([3, np.nan, 2, None])
```

```
Out(10): 0    1.0
1     NaN
2     2.0
3     NaN
```

```
dtype: float64
```

17. Define the methods used in null values.

There are several useful methods for detecting, removing, and replacing nu values in Pandas data structures. They are:

isnull()-Generate a Boolean mask indicating missing values

`notnull()`-Opposite of `isnull()`

`dropna()`-Return a filtered version of the data

`fillna()`-Return a copy of the data with missing values filled or imputed

18. Name the categories of Join.

The `pd.merge()` function implements a number of types of joins: the one-to-one, many-to-one, and many-to-many joins.

19. Mention the term GroupBy and the steps involved in it.

The name "group by" comes from a command in the SQL database language, but it is perhaps more illuminative to think of it in the terms first coined by Hadley, Wickham of Rstats fame: split, apply, combine.

The split step involves breaking up and grouping a DataFrame depending on the value of the specified key

apply to involves computing some function, usually an aggregate Transformation, or filtering, within the individual groups. The combine step merges the results of these operations into an output array.

20. Define Pivot Table

A pivot table is a statistical operation that is commonly seen in spreadsheets and other programs that operate on tabular data. The pivot table takes simple column wise data as input, and groups the entries into a two-dimensional table that provides a multidimensional summarization of the data.

21. Define Hierarchical Indexing.

Hierarchical Indexes are also known as multi-indexing is setting more than one column name as the index.

PART B QUESTIONS

1. Explain about the Datatypes in Python?
2. Explain about the Basics of NumPy arrays with coding?
3. Describe the syntax accessing sub arrays in an arrays using array Slicing?
4. Explain in detail about Reshaping of arrays?
5. Explain about the Aggregation operation in Python?
6. Explain about broadcasting and its rules with examples?
7. Explain about Boolean masking in detail?
8. Explain in detail about Fancy indexing?
9. Explain in detail why there is a need for NumPy's structured arrays?
10. Describe the installation Procedure and using of Pandas?
11. Explain the different ways of constructing a Pandas data frame?
12. Explain about the data selection in series and data frames?
13. Explain about the different operations on data in Pandas?
14. Explain how handling of missing data is done?

15. Explain about the methods for detecting, removing and replacing Null values in Pandas data structures.
16. Explain in detail about hierarchical indexing?
17. Describe data aggregations on Multiindices?
18. Explain in detail about combining datasets using Concat and Append?
19. Explain in detail about categories of Joins.
20. Explain about the steps involved in GroupBy with suitable diagram and coding?

UNIT III

MACHINE LEARNING

PART A QUESTIONS AND ANSWERS

1. What is Machine Learning?

Machine learning is a subfield of artificial intelligence that involves the development of algorithms and statistical models that enable improve their performance in tasks through experience. A computer is said to learn from task T and improve its Performance(P) from experience E. computers

2. What are the steps defined in modelling phase. The modeling phase consists of four steps:

- Feature engineering and model selection
- Training the model
- Model validation and selection
- Applying the trained model to unseen data

3. How machine learning is different from general programming?

In general programming, we have the data and the logic by using these two we we have the data and the and we het the machine learn the logic from them so, that the same logic can be used to answer the questions which will be faced in the future. Also, there are times when writing logic in codes is not possible so, at those times machine learning becomes a saviour and learns the logic itself.

4. How to train a model in machine learning?

Three steps to training a machine learning model

Step 1: Begin with existing data.Learning

Step 2: Analyze data to identify patterns.

Step 3: Make predictions.

5. What are the error measures in machine learning?

Two common error measures in machine learning are the

(i) **classification** error rate for classification problems and squared error for regression

(ii) **the mean problems**

6. Mention the validation strategies available in Data Science

Dividing your data into a training set with X% of the observations and keeping the rest as a holdout data set.

K-folds cross validation: This strategy divides the data set into k parts and uses each part one time as a test data set while using the others as a training data set.

Leave-1 out: This approach is the same as k-folds but with k=1. Always leave one observation out and train on the rest of the data.

7. What are the types of machine learning?

The two big types of machine learning techniques

1. Supervised: Learning that requires labeled data

2. Unsupervised: Learning that doesn't require labeled data but is usually less accurate or reliable than supervised learning.

3. Semi-supervised learning is in between those techniques and is used when only a small portion of the data is labeled.

8. What is the Classification Algorithm?

The Classification algorithm is a Supervised Learning technique that is used to identify the category of new observations on the basis of training data. In Classification, a program learns from the given dataset or observations and then classifies new observation into a number of classes or groups such as, Yes or No, 0 or 1, Spam or Not Spam, cat or dog, etc. Classes can be called as targets/labels or categories.

9. Define Regression.

A measure of the relation between the mean value of one variable A corresponding values of other variables

10. Define regression line.

The regression line is a straight line rather than a curved line that denotes linear relationship between the two variables.

11. What is a predictive error?

Predictive error refers to the difference between the predicted values made by some model and the actual values. Example predictive error denotes the assumption of number of cards received and the actual cards received.

12. What is meant by Captcha Check?

What is were hard numbers that the human user must decipher and enter into a form field before sending the form back to the web server.

13. Define Clustering.

Clustering is the task of dividing the unlabeled data or data points into different clusters such that similar data points fall in the same cluster than those which differ from the others.

14. Define Principal Component Analysis Principal Component Analysis

A technique to find the latent variables in the data set while retaining as much information as possible.

15. Define Semi Supervised Learning

Semi-supervised learning is a type of machine learning that falls in between supervised and unsupervised learning. It is a method that uses a small amount of labeled data and a large amount of unlabeled data to train a model. The goal of semi-supervised learning is to learn a function that can accurately predict the output variable based on the input variables, similar to supervised learning.

16.State the learners in classification problem. Mention about Lazy Learners.

Lazy Learners: Lazy Learner firstly stores the training dataset and wait until receives the test dataset. In Lazy learner case, classification is done on the ba of the most related data stored in the training dataset. It takes less time in train but more time for predictions.

17.Mention the Types of ML Classification Algorithm:

Linear Models

- Logistic Regression
- Support vector machines

Non-linear Models

- Support Vector Machines
- K-Nearest Neighbours
- Kernel SVM
- Naïve Bayes
- Decision Tree Classification
- Random Forest Classification

18. State Outlier analysis.

Outlier is data object that deviates significantly from the rest of the data objects and behaves from un different manner. An outlier the rest of the data objects significantly from the rest of the objects. They can be caused by measurement or execution errors. The analysis of outlier data is referred to as outlier analysis or outlier mining.

19. State the difference between Supervised and Unsupervised Learning

Parameter

- Supervised Learning
- Unsupervised Learning

Input Data

- Uses Known and Labeled Data as input
- Uses Unknown Data as input

Computational Complexity

- Less Computational Complexity
- More Computational Complex

Real Time

- Uses off-line analysis
- Uses Real Time Analysis of Data

Number of Classes

- Number of Classes are known
- Number of Classes are not known

Accuracy of Results

- Accurate and Reliable Results
- Moderate Accurate and
- And Reliable Results.

Output data

- Desired output is given.
- Desired output is not given.

Model

- In supervised learning it is not possible to learn larger and more complex models than with supervised learning
- In supervised learning it is not possible to learn larger and more complex models than with supervised learning

Training data

- In supervised learning training data is used to infer model
- In unsupervised learning training data is not used.

20. What are the steps involved in Data Science Process.

1. Setting the research goal
2. Retrieving Data
3. Data Preparation
4. Data Exploration
5. Data Modelling
6. Presentation and Automation

21. What is a Multi-class Classifier?

- Multi-class Classifier: If a classification problem has more than two outcomes, then it is called as Multi-class Classifier.

Example: Classifications of types of crops, Classification of types of music.

22. Mention the clustering techniques used in performing tasks.

The clustering technique can be widely used in various tasks. Some most common uses of this technique are:

- Market Segmentation
- Statistical data analysis
- Social network analysis
- Image segmentation
- Anomaly detection

23. What is meant by K-means algorithm?

K-Means algorithm: The k-means algorithm is one of the most popular clustering algorithms. It classifies the dataset by dividing the samples into different clusters of equal variances. The number of clusters must be specified in this algorithm. It is fast with fewer computations required, with the linear complexity of $O(n)$

24. State DBSCAN algorithm

DBSCAN Algorithm: It stands for Density-Based Spatial Clustering of Applications with Noise. It is an example of a density-based model similar to the but with some are separated by the areas of low density. Because of this, the high density clusters can be found in any arbitrary shape. Example: K-NN algorithm, Case-based reasoning

PART B QUESTIONS

1. Explain the steps involved in modeling process.
2. Explain the types of Machine learning.
3. Explain about Supervise learning with suitable example.
4. Explain in detail about unsupervised learning with an example.
5. Explain the steps involved in discerning digits from images under supervised learning.
6. Explain in detail about unsupervised learning for finding latent variables in a wine quality data set.
7. Explain about Semi Supervised Learning in detail.
8. Explain Classification Algorithm in Machine Learning
9. Explain about main clustering methods used in Machine Learning.
10. Define regression. Explain about prediction of values using the regression line
11. Write short notes on (a) Clustering Algorithms (b) Applications of Clustering techniques
12. Explain in detail about Outlier analysis.

UNIT IV

DATA VISUALIZATION

PART A QUESTIONS AND ANSWERS

1. Define Matplotlib.

Matplotlib is a cross-platform, data visualization and graphical plotting library for Python and its numerical extension NumPy.

2. How to import Matplotlib?

matplotlib.pyplot is a collection of command style functions that make matplotlib work like MATLAB. Each pyplot function takes one command style funding to a figure: eg., creates a figure, creates a plotting area in a figure, plots some lines in a plotting area, decorates the plot with labels, etc. In matplotlib.pyplot various states are preserved across function calls, so that it keeps track of things like the current figure and plotting area, and the plotting functions are directed to the current axes.

```
import matplotlib.pyplot as plt
plt.plot([1,2,3,4])
```

```
plt.ylabel('some numbers')
```

```
plt.show()
```

3. How to specify line colors?

A plot is used to control the line colors and styles. The plt.plot() function takes additional arguments that can be used to specify these. To adjust the color, the color keyword is used, which accepts a string argument representing virtually any imaginable color. The color can be specified in a variety of ways.

In [6]:

```
plt.plot(x, np.sin(x0), color='blue') # specify color by name plt.plot(x, np.sin(x - 1), color='g') # short  
color code (rgbcmyk)
```

```
plt.plot(x, np.sin(x - 2), color='0.75') # Grayscale between and 1
```

```
plt.plot(x, np.sin(x3), color='#FFDD44') # Hex code (RRGGBB from 00 to FF)
```

```
plt.plot(x, np.sin(x4), color=(1.0,0.2,0.3)) # RGB tuple,
```

values 0 and 1

```
plt.plot(x, np.sin(x5), color='chartreuse'); # all HTML color names supported
```

4. Define Scatter plots.

Instead of points being joined by line segments, here in scatter plots the points are represented individually with a dot, circle, or other shape

5. Mention the difference between plt.scatter and plt.plot.

The primary difference of plt.scatter from plt.plot is that it can be used to create scatter plots where the properties of each individual point (size, face color, edge color, etc.) can be individually controlled or mapped to data whereas the plt.plot function only draws a line from point to point.

6. Define Contour Plot.

A contour plot is a graphical technique for representing a 3-dimensional surface by plotting constant z slices, called contours, on a 2-dimensional format. A contour plot can be created with the plt.contour function.

It takes three arguments:

- a grid of x values,
- a grid of y values, and
- a grid of z values.

The x and y values represent positions on the plot, and the z values will be represented by the contour levels.

7. Define Kernel Density Estimation.

One of the common methods of evaluating densities in multiple dimensions is Kernel Density Estimation (KDE). KDE can be thought of as a way to "smear out" the points in space and add up the result to obtain a smooth function.

8. Define Plot Legends.

Plot legends give meaning to a visualization, assigning labels to the various plot elements. The simplest legend can be created with the plt.legend() command, which automatically creates a legend for any labeled plot elements.

9. State the categories of colormaps

There are three different categories of colormaps

- Sequential colormaps-These consist of one continuous sequence of colors (e.g., binary or viridis).
- Divergent colormaps-These usually contain two distinct colors, which show positive and negative deviations from a mean (eg, RdBu or PuOr)
- Qualitative colormaps-These mix colors with no particular sequence (eg rainbow or jet).

10. What are Subplots?

Matplotlib has the concept of subplots: groups of smaller axes that can exist together within a single figure. These subplots might be insets, grids of plots exist together complicated layouts. The most basic method of creating an axes is to use the `plt.axes` function, `plt.subplot()`, which creates a single subplot within a grid.

11. What is a major and minor ticks?

Within each axis, there is the concept of a major tick mark and a minor tick mark. As the names would imply, major ticks are usually bigger or more pronounced, while minor ticks are usually smaller.

12. What is the need for visualizing errors?

For any scientific measurement, accurate accounting for errors is nearly as important, if not more important, than accurate reporting of the number itself.

In visualization of data and results, showing the errors effectively can make a plot convey much more complete information.

13. What are the types of transforms present?

There are three predefined transforms that can be useful in this situation:

`ax.transData`-Transform associated with data coordinates

`ax.transAxes`-Transform associated with the axes (in units of axes dimensions)

`fig.transFigure`-Transform associated with the figure (in units of figure dimensions)

14. How hiding tick/labels is done?

The most common tick/label formatting operation is the act of hiding ticks or labels. This is done by using `plt.NullLocator()` and `plt.NullFormatter()`.

15. What is a surface plot?

A surface plot is like a wireframe plot, but each face of the wireframe is a filled polygon. Adding a colormap to the filled polygons can aid perception of the topology of the surface being visualized.

16. What is meant by Geographic Data with Basemap?

One common type of visualization in data science is that of geographic data. Matplotlib's main tool for this type of visualization is the Basemap toolkit, which is one of several Matplotlib toolkits that live under the `mpl_toolkits` namespace. Admittedly, Basemap feels a bit clunky to use, and often even simple visualizations take much longer to render. More modern solutions, such as leaflet or the Google Maps API, may be a better choice for more intensive map visualizations. Still, Basemap is a useful tool for Python users to have in their virtual toolbelts.

17. Define cylindrical projections.

The simplest of map projections are cylindrical projections, in which lines of constant latitude and longitude are mapped to horizontal and vertical lines, respectively. This type of mapping represents equatorial regions quite well, but results in extreme distortions near the poles.

PART B QUESTIONS

1. Explain about Matplotlib with its import, setting styles and displaying the plots.
2. Explain about the dual interface of Matplotlib.

3. Explain in detail about simple Line plots with line colors, styles and axes limits.
4. Explain in detail about simple scatter plots.
5. Explain how Visualizing Errors is done.
6. Explain in detail about Density and Contour Plots.
7. Explain in detail about histogram functions with Binnings and Density.
8. Explain in detail about Customizing Plot Legends.
9. Explain in detail about Customizing Colorbars.
10. Explain in detail about Multiple Subplots.
11. Describe the example Effect of Holidays on US Births with Text and Annotation.
12. Explain in detail about Customizing Ticks.
13. Explain in detail the various built in styles in matplotlib stylesheets.
14. Explain in detail about Three-Dimensional Plotting ion Matplotlib.
15. Explain in detail about Visualizing a Mobius Strip.
16. Explain about Geographic Data with Basemap with different Map Projections, Map Background and Plotting Data on Maps.
17. Explain the example California Cities.
18. Explain the example Surface Temperature Data.
19. Explain in detail about Visualization with Seaborn.

UNIT V

HANDLING LARGE DATA

PART A QUESTIONS AND ANSWERS

1. **What are the general problems you face while handling large data managing massive amounts of data. It's in the name big data is big Integrating data from multiple sources.**
 - Ensuring data quality.
 - Keeping data secure.
 - Selecting the right big data tools.
 - Scaling systems and costs efficiently.
 - Lack of skilled data professionals.
 - Organizational resistance.
2. **What are the problems encountered when working with more data in memory.**
 - Not enough memory
 - Processes that never end
 - Some components from a bottleneck while others remain idle
 - Not enough speed
3. **Mention the solutions for handling large data sets.**
 - Choose the right algorithms
 - Choose the right data structures
 - Choose the right tools

4. What are the algorithms involved in handling large data se

- Online algorithms
- Block Matrices
- MapReduce

5. Define Perceptron

A perceptron is one of the least complex machine learning binary classification (0 or 1)

6. State how the `train_observation()` function works.

This function has two large parts.

- The first is to calculate the prediction of an observation and compare it to the actual value.
- The second part is to change the weights if the prediction seems to be wrong.

7. What are the options available for online algorithms learning.

There are three options:

1. **Full batch learning (also called statistical learning)** - Feed the algorithm all the data at once.
2. **Mini-batch learning** Feed the algorithm a spoonful (100, 1000, depending on what your hardware can handle) of observations at a time.
3. **Online learning** - Feed the algorithm one observation at a time.

8. What is meant by Mapreduce algorithms?

MapReduce implements various mathematical algorithms to divide a task into small parts and assign them to multiple systems.

Ex. To count all the votes for the national elections. The country has 25 parties, 1,500 voting offices, and 2 million people. Choose to gather all the voting tickets from every office individually and count them centrally, or ask the local offices to count the votes for the 25 parties and hand over the results, and could then aggregate them by party

9. What do you mean by CRUD?

CRUD is the acronym for CREATE, READ, UPDATE and DELETE. These terms describe the four essential operations for creating and managing persistent data elements, mainly in relational and NoSQL databases

10. What is Sparse Data?

Sparse data is a variable in which the cells do not contain actual data within data analysis. Sparse data is empty or has a zero value. Sparse data is different from missing data because sparse data shows up as empty or zero while missing data doesn't show what some or any of the values are. LM

11. When sparse data is present in your datasets, it creates dense data

There are two types of sparsity:

1. Controlled sparsity happens when there is a range of values for multiple dimensions that have no value
2. Random sparsity occurs when there is sparse data scattered randomly throughout the datasets

12. State the different Python Tools

Python Tools

Python has a number of libraries that can help to deal with large data. They range from smarter data structures over code optimizers to just-in-time compilers.

The following is a list of libraries we like to use when confronted with large data:

- Cython
- Numexpr
- Numba
- Bcolz
- Blaze
- Theano
- Dask

13. Mention the three general programming tips for dealing with large data sets.

1. Don't reinvent the wheel.
2. Get the most out of your hardware.
3. Reduce the computing need.

14. What is meant by Predicting Malicious URLs?

Predicting malicious URLs is essential for protecting users and organizations from a wide range of cyber threats. It is a dynamic field that requires continuous adaptation to evolving threats, making use of a combination of methods and technologies to ensure robust cybersecurity.

15. How building a recommender system is done?

Building a recommender system inside a database leverages the strengths of a Building a recommender recommendation capabilitiesent dalading real-time recommendations, scalability, personalization, and efficient data management. It can be particularly useful in scenarios where low-latency and high-performance recommendations are critical, such as e-commerce platforms, streaming services, and content recommendation systems.

16. What is the main purpose of Locality-Sensitive Hashing

A technique is used to ensure similar customers (local optima) without guaranteeing that it finds the best customer (global optimum). A common technique used to solve this is called Locality-Sensitive Hashing. The idea behind Locality-Sensitive Hashing is simple: Construct functions that map similar customers close together (they're put in a bucket with the same label) and make sure that objects that are different are put in different buckets.

17. What is meant by hamming distance?

The distance that used is to compare customers is called the hamming distance. The hamming distance is used to calculate how much two strings differ. The distance is defined as the number of different characters in a string.

18. How does MYSQL database connect to Python library?

MySQL database connection Python library To connect to this server from Python there is a need to install SQL Alchemy or another library capable of communicating with MySQL. Use MySQLdb and on Windows use Conda right off the bat to install it.

19. Define SVMLight.

SVMLight is a machine learning software package developed by Thorsten Joachims. It is designed for solving classification and regression problems using Support Vector Machines (SVMs), which are a type of supervised learning algorithm.

20. What is a Tree Structure?

Trees are a class of data structure that allows to retrieve information much faster than scanning through a table. A tree always has a root value and subtrees of children, each with its children, and so on. Simple examples would be a family tree or a biological tree and the way it splits into branches, twigs, and leaves.

PART B QUESTIONS

1. Explain in detail about the problems that you face when handling large data.
2. Explain in detail about the general techniques for handling large volumes of data.
3. Explain the steps involved in training the perceptron by observation.
4. Explain the steps involved in train functions
5. Explain in detail about block matrix calculations with bcolz and Dask libraries.
6. Explain in detail about general programming tips for dealing with large data sets.
7. Explain the case study Predicting Malicious URLs.
8. Explain the steps for creating a model to distinguish the malicious from the normal URLs.
9. Explain in detail about Building a recommender system inside a database.
10. Explain in detail the steps involved in creating the hamming distance.
11. Explain about the customers in the database who watched the movie or not using a recommender system.

*||| **ALL THE BEST** |||*